

Lithium-Ion Battery Charger

EFY LAB

Like most people, you probably have some discarded devices like MP3 players and cell phones lying around that use lithium-ion batteries for power supply. Usually their electronics cannot be used for any other purpose, but their batteries can still be used, even for toys. Since electronics hobbyists tend to be an inventive lot, they will usually find a way to incorporate and recycle a battery. However, you need to charge these batteries.

Some original equipment includes charger circuit for the battery, most likely on a small part of the PCB. It is difficult to figure out which components form part of the charger circuit since no circuit diagram is normally available. In that case, we have to build our own lithium-ion charger as explained in this article.

Circuit and working

The charger circuit described here is built around lithium-ion charger IC MAX8677A (Fig. 1). This IC works completely autonomously, so there

is no need of a microcontroller (and hence no software). LEDs are used by the IC to indicate state of the charging process.

MAX8677A is very flexible and has three electronic switches to control the charge and load currents according to the situation. With an external power source, the IC can use the available power to both charge the battery and supply the load. If the load requires more power than the charger can deliver, MAX8677A can make the battery provide the extra current. When there is no external power source available, the load is powered solely by the battery.

The IC can be powered from a USB port via pins 15 and 16 (USB). In that case, the current drawn is limited to 500mA (the maximum for a USB 2.0 port). USB port pins are connected to ground if not used. The IC can also be powered from an adaptor via pins 2 and 3 (DC), where the current limit can be increased up to a maximum of 2A.

For the charger circuit, you can use the DC input, which gives us greater flexibility in setting various limits. Op-

erating voltage at the input is between 4.1V and 6.6V. If the voltage becomes too high, MAX8677A turns off the input to prevent it from overheating. The IC can survive voltage spikes up to a maximum of 14V. The charging status is indicated by the LEDs: the battery is being charged (LED3), the battery is fully charged (LED1), the battery is faulty (LED2).

Maximum charge current and maximum input current can be set with this IC; the second value should always be larger than the first. If this is not the case, the programmed maximum charge current can never be reached, as it cannot go higher than the maximum input current. Both these limits are set using a resistor.

Maximum charge current:

$$ICHGMAX = 3000 / R1SET = 3000 / R9 = 3000 / 5.6k\Omega = 535mA$$

Maximum input current:

$$IDCMAX = 3000 / RPSET = 3000 / R6 = 3000 / 3.3k\Omega = 909mA$$

PARTS LIST

Semiconductors:

- IC1 - MAX8677A charger
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R5 - 4.7-kilo-ohm
- R3, R4 - 560-kilo-ohm
- R6 - 3.3-kilo-ohm
- R7, R8 - 10-kilo-ohm
- R9 - 5.6-kilo-ohm

Capacitors:

- C1, C3 - 4.7 μ F, 16V electrolytic
- C2 - 0.1 μ F ceramic disk
- C4 - 0.068 μ F ceramic disk

Miscellaneous:

- CON1 - Mini USB connector
- CON2 - 2-pin terminal connector
- CON3, CON4 - 2-pin connector
- F1 - 500mA fuse
- F1 - 3.7V Li-ion battery

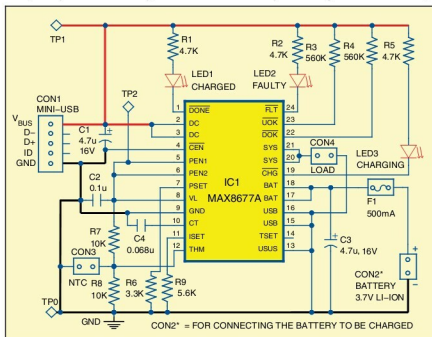


Fig. 1: Circuit diagram of the lithium-ion battery charger

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	3.3V